

Security Levels

The security parameters considered in our implementation are consistent with the security levels adopted by the SQIsign2D² parameter sets. In particular, we distinguish between the “eff” and “sec” parameter sets at each security level. The former targets the quantum security of the scheme, whereas the latter provides a more conservative parameter selection with respect to the classical security level.

More precisely, the security levels used in our experiments are summarized as follows:

- **Level 2:** The `Level2-eff` and `Level2-sec` parameter sets target approximately 80-bit quantum security and 128-bit classical security.
- **Level 3:** The `Level3-eff` and `Level3-sec` parameter sets target approximately 128-bit quantum security and 256-bit classical security.
- **Level 5:** The `Level5-eff` and `Level5-sec` parameter sets target approximately 256-bit quantum security and 512-bit classical security.

Thus, the security levels can be summarized by

Security Level	Classical Security	Quantum Security
Level 2	128-bit	80-bit
Level 3	256-bit	128-bit
Level 5	512-bit	256-bit

The distinction between the `eff` and `sec` parameter sets does not correspond to different target security categories. Rather, both parameter sets belong to the same security level and are designed to meet the same classical and quantum security targets, while using different parameter choices that trade off efficiency and security margins. In particular, `eff` denotes the efficiency-oriented parameter set, whereas `sec` denotes the security-oriented parameter set.

Accordingly, the performance results in Table 6.1 and Table 6.2 should be interpreted within these three security categories. Level 2 corresponds to the approximately 128-bit classical security category, Level 3 corresponds to the approximately 256-bit classical security category, and Level 5 corresponds to the approximately 512-bit classical security category. The corresponding quantum security targets are approximately 80, 128, and 256 bits, respectively.